

VALDEZ AP, AK, USA

WMO: 702756

Lat: 61.132N

Lon: 146.244W

Elev: 18

StdP: 101.11

Time Zone: -9.00 (NAK)

Period: 96-19

WBAN: 26479

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.0	-14.1	-27.7	0.3	-10.6	-25.9	0.4	-9.1	14.6	-4.8	11.9	-2.9	2.3	90	0.680

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.2	21.3	13.5	19.0	13.0	17.6	12.3	14.7	19.4	13.8	18.1	13.0	16.9	3.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.4	9.0	16.3	11.9	8.7	15.5	11.1	8.2	14.6	41.0	19.4	38.5	18.1	36.5	16.8	17.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.8	7.7	5.6	-20.0	24.7	2.1	2.7	-21.5	26.7	-22.8	28.3	-24.0	29.8	-25.5	31.8		
			-20.6	15.7	2.3	1.0	-22.2	16.4	-23.6	17.0	-24.9	17.5	-26.6	18.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.5	-5.1	-3.6	-2.0	2.8	7.8	11.2	12.5	11.7	8.6	3.9	-2.1	-4.3	(6)
(7)		DBStd	7.28	5.19	4.25	3.63	2.65	2.83	2.15	1.88	1.94	2.10	2.80	4.31	4.52	(7)
(8)		HDD10.0	2577	469	380	373	216	81	9	1	3	51	189	364	442	(8)
(9)		HDD18.3	5422	727	613	631	465	327	213	182	208	293	447	614	701	(9)
(10)		CDD10.0	198	0	0	0	1	12	46	78	54	8	0	0	0	(10)
(11)		CDD18.3	1	0	0	0	0	0	0	0	1	0	0	0	0	(11)
(12)		CDH23.3	20	0	0	0	0	0	8	4	8	0	0	0	0	(12)
(13)		CDH26.7	2	0	0	0	0	0	1	0	1	0	0	0	0	(13)
(14)	Wind	WSAvg	1.6	1.5	1.5	2.1	1.4	2.0	2.0	1.6	1.3	1.2	1.3	1.7	1.4	(14)
(15)	Precipitation	PrecAvg	1673	150	142	112	88	75	70	93	163	221	211	144	182	(15)
(16)		PrecMax	2370	386	361	254	206	169	154	228	463	424	392	523	440	(16)
(17)		PrecMin	884	0	14	21	14	18	3	5	9	18	8	11	34	(17)
(18)		PrecStd	325	96	87	67	51	35	35	48	98	99	101	109	109	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.1	5.2	7.5	13.9	20.2	23.9	23.2	24.0	20.6	12.5	7.6	4.2	(19)
(20)		MCWB	1.2	2.0	2.3	7.6	10.9	14.0	15.4	13.8	11.1	5.8	3.5	1.6	(20)	
(21)		2%	DB	2.8	3.0	5.2	11.1	17.8	20.9	21.0	20.7	16.3	11.0	5.2	2.8	(21)
(22)		MCWB	1.1	0.7	1.6	5.4	9.8	13.5	14.8	13.1	9.9	5.6	2.6	1.5	(22)	
(23)		5%	DB	2.2	2.4	3.8	8.7	16.0	17.9	18.7	17.9	13.9	8.9	3.7	2.2	(23)
(24)		MCWB	1.0	0.6	0.8	4.1	8.9	12.2	14.2	12.9	9.3	5.3	1.6	1.1	(24)	
(25)		10%	DB	1.2	1.4	2.7	7.5	13.0	17.0	17.3	16.4	12.5	7.8	2.6	1.2	(25)
(26)		MCWB	0.4	0.2	0.3	3.7	7.9	11.7	13.1	12.1	9.1	4.8	0.8	0.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.0	2.8	4.1	7.9	12.0	15.3	16.3	15.6	12.0	8.5	5.0	3.0	(27)
(28)		MDCB	4.1	4.0	6.6	13.8	18.2	22.0	21.1	20.4	16.0	9.7	7.0	3.3	(28)	
(29)		2%	WB	1.9	1.8	2.7	5.8	10.5	13.7	15.1	14.3	11.2	7.6	3.4	2.0	(29)
(30)		MDCB	2.3	2.6	4.2	9.4	16.3	19.3	19.5	18.7	14.5	8.5	4.6	2.4	(30)	
(31)		5%	WB	1.2	1.2	1.8	4.8	9.5	12.7	14.1	13.4	10.5	6.7	2.4	1.2	(31)
(32)		MDCB	1.6	1.8	2.9	7.8	14.2	17.7	18.2	17.4	13.2	7.9	3.3	1.6	(32)	
(33)		10%	WB	0.4	0.5	1.0	4.0	8.6	11.8	13.3	12.5	9.9	5.8	1.3	0.5	(33)
(34)		MDCB	1.0	1.2	2.2	7.0	13.2	16.1	17.0	15.6	12.2	7.4	2.2	1.0	(34)	
(35)	Mean Daily Temperature Range	MDBR	5.3	5.8	6.0	6.3	7.2	7.7	7.2	7.2	6.5	5.3	4.7	5.0	(35)	
(36)		5% DB	MCDBR	5.4	5.9	7.0	9.5	12.3	12.4	11.5	11.4	10.3	7.7	5.3	5.4	(36)
(37)		MCWBR	4.2	4.5	4.9	5.8	6.5	7.2	7.3	7.1	6.2	4.8	3.6	4.1	(37)	
(38)		5% WB	MCDBR	4.2	4.4	5.2	7.8	10.1	11.0	9.9	9.7	7.6	5.2	4.0	4.1	(38)
(39)		MCWBR	3.8	3.8	3.9	5.1	5.8	6.8	6.8	6.8	5.3	4.1	3.2	3.6	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.257	0.315	0.322	0.386	0.400	0.406	0.402	0.401	0.343	0.318	0.270	0.247	(40)	
(41)		taud	2.010	2.035	2.075	1.940	2.051	2.163	2.238	2.244	2.373	2.196	1.988	1.897	(41)	
(42)		Ebn at Noon	437	623	777	786	814	818	811	777	767	626	395	253	(42)	
(43)		Edh at Noon	42	80	107	150	146	134	121	111	81	67	42	29	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.28	0.99	2.58	4.40	5.46	5.63	5.04	3.84	2.47	1.13	0.41	0.15	(44)	
(45)		RadStd	0.03	0.14	0.24	0.33	0.37	0.33	0.31	0.37	0.32	0.13	0.06	0.01	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page